National 5 Engineering Science – Energy Revision Sheet

Key Formulae

- **Work Done:**
  $W = F \times d$
  (Work = Force × distance moved in the direction of the force)
- **Kinetic Energy:**
  $E_k = \frac{1}{2}mv^2$
- **Potential Energy:**
  $E_p = mgh$
- **Heat Energy:**
  $Q = mc\Delta T$
  (m = mass, c = specific heat capacity, ΔT = temperature change)
- **Electrical Energy:**
  $E = VIt$
  (V = voltage, I = current, t = time)
- **Efficiency:**
  η = (Useful Energy Out ÷ Total Energy In) × 100%

Energy Types

Energy Type	Description	Example
Kinetic	Energy of movement	Moving car, thrown ball
Potential	Stored energy due to position	Raised mass, water in a dam
Heat (Thermal)	Energy related to temperature; often wasted	Friction, boilers, hot components
Electrical	Energy from moving charges (electricity)	Mains supply, batteries
Chemical	Energy stored in chemical bonds	Petrol, food, batteries

Energy Transformations & Losses

- Energy can change from one form to another (e.g., electrical → heat in a kettle).
- **Useful energy** is the energy output we want.
- **Wasted energy** is often heat or sound.
- Example: Car engine – Chemical → Kinetic (useful), Heat & Sound (wasted).

Efficiency

- Efficiency = (useful energy out ÷ total energy in) × 100%.
- Can be improved by insulation, lubrication, and better design.

Energy Audits

- A systematic check of where energy is used and wasted.
- Measures consumption, identifies inefficient equipment, and recommends improvements (e.g., LED lighting, improved insulation, switching off unused equipment).

Quick Tips for the Exam

- Always state units (Joules, Watts, %).
- Show working clearly in calculations.
- Identify both useful and wasted energy when describing systems.

- The number of significant figures expressed in a final answer should be equivalent to the least significant data value given in the question. For example,

  A crate of mass 12.34 kg (4 sig figs) is lifted to a height of 8.55 m (3 sig figs) on a platform. Take g = 9.8 m/s² (2 sig figs). Calculate the gravitational potential energy of the crate.

  Step 1: Complete the calculation with no rounding.

  Step 2: Determine significant figures in the question.

  Mass: 12.34 → 4 sig figs

  Gravity: 9.8 → 2 sig figs

  Height: 8.55 → 3 sig figs

  Least precise = 2 sig figs (gravity) → **final answer must have 2 significant figures**